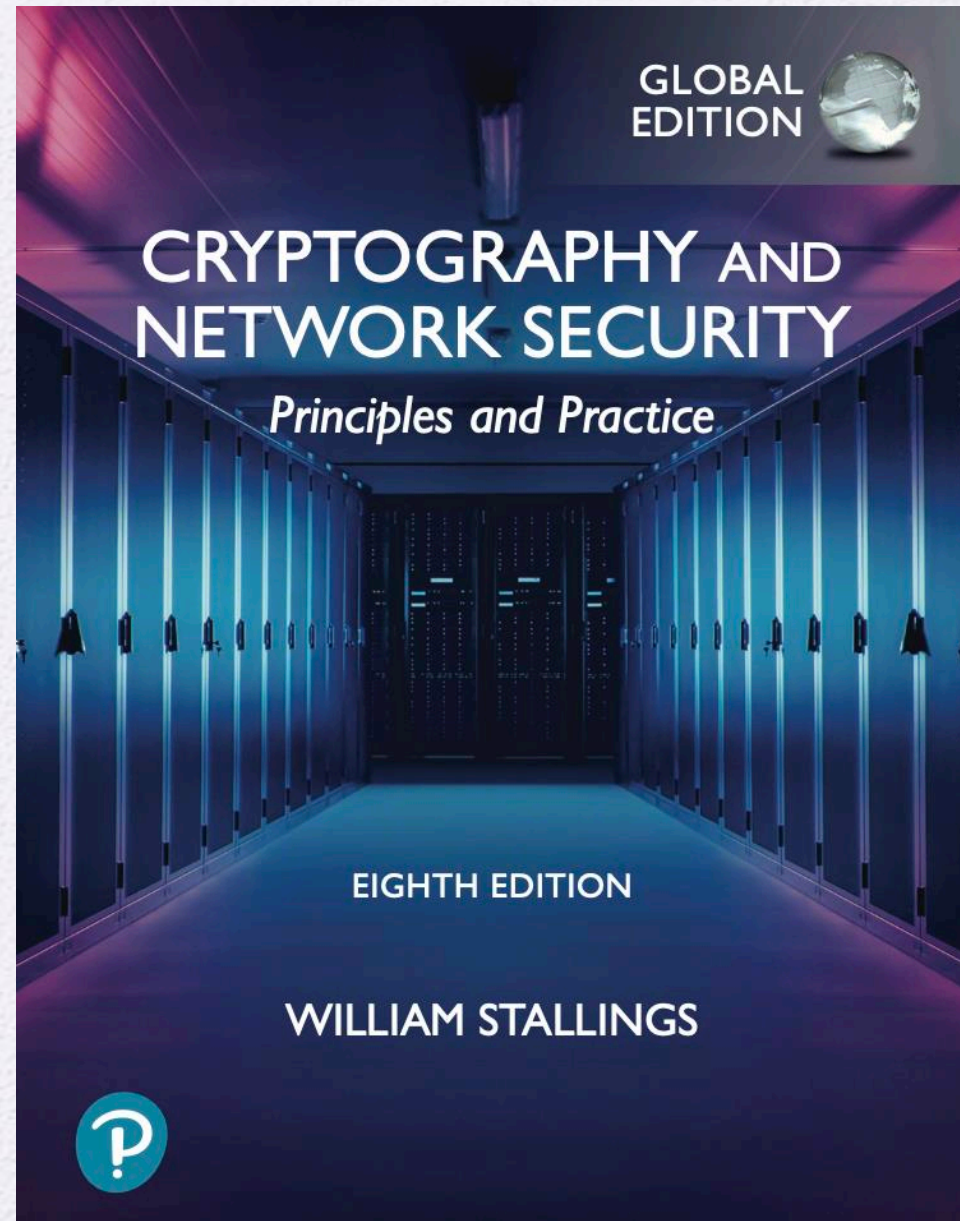


University of Nevada – Reno
Computer Science & Engineering
Department

CS454/654 Reliability and Security
of Computing Systems - Fall 2024



NETWORK ENDPOINT SECURITY

21.1 Firewalls

- Firewall Characteristics
- Types of Firewalls
- DMZ Networks

21.2 Intrusion Detection Systems

- Basic Principles
- Approaches to Intrusion Detection
- Host-Based Intrusion Detection Techniques
- Network-Based Intrusion Detection Systems

21.3 Malicious Software

- Types of Malware
- Malware Defense

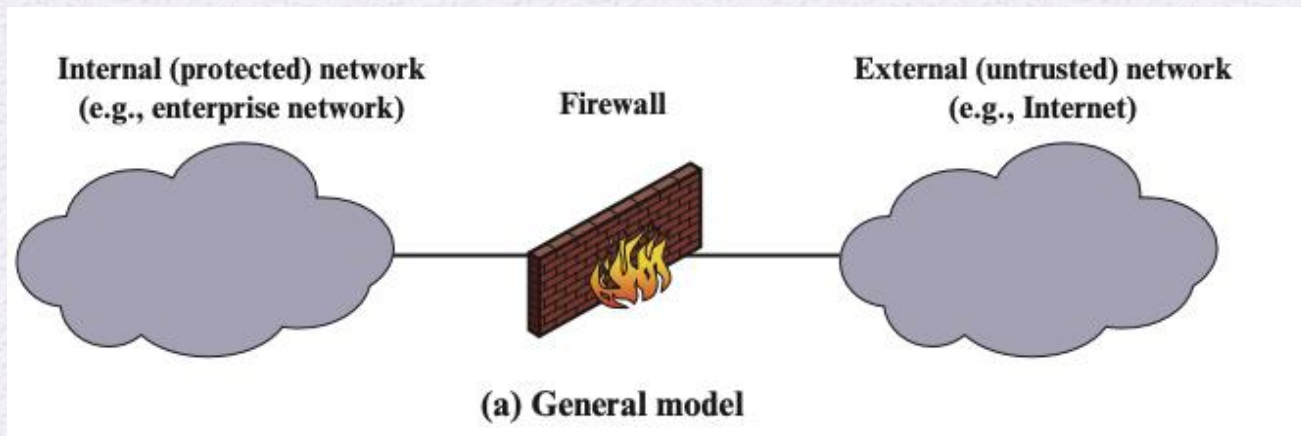
21.4 Distributed Denial of Service Attacks

- DDoS Attack Description
- Constructing the Attack Network
- DDoS Countermeasures

21.5 Key Terms, Review Questions, and Problems

Firewalls

- **Deployed** on **outer perimeter** of the network infrastructure
- Also deployed in **inner** network to **segregate portions** of the network.
- Enforces access policy.



- **Four techniques** that firewalls use to control access
 - **Service control:** What type of services can be accessed
 - **Direction control:** What should be the direction of the traffic for service
 - **User control:** Tailoring the access depending on user.
 - **Behavioral control:** How particular service can be used.

Firewalls

Firewall enables

- Traffic **monitoring**.
- Network address translation (**NAT**) and logging internet usage.
- Facilitate Virtual Private Networks (**VPN**)

However, firewall **can not protect internal threats**.

Types of Firewalls

- Packet Filtering
- Stateful Inspection Firewalls
- Application Level Gateway
- Circuit-level gateway

Packet Filtering

- Apply filtering to traffic based on source/destination IP, protocol, and port number.
- Examine each individual packet and **apply predefined rules**.
- Cisco ASA 5500-X Series Firewalls - Supports simple packet filtering
<https://www.cisco.com/c/en/us/support/security/asa-5500-series-next-generation-firewalls/series.html>

Rule Set A

action	Ourhost	port	theirhost	port	comment
block	*	*	SPIGOT	*	we don't trust these people
allow	OUR-GW	25	*	*	connection to our SMTP port

Rule Set B

action	Ourhost	port	theirhost	port	comment
block	*	*	*	*	default

Rule Set C

action	Ourhost	port	theirhost	port	comment
allow	*	*	*	25	connection to their SMTP port

Packet Filtering Attacks & Countermeasures

1. IP address spoofing. Spoofed IP address belongs to an internal host.

Countermeasure: discard packets with an inside source IP address if the packet arrives on an external interface (typically implemented on a router external to the firewall).

2. Source Routing Attacks. It attempts to specify the route (path on the Internet) in hopes that it will bypass security measures that do not analyze the source routing information.

Countermeasure: discard all packets that use this option.

3. Tiny fragment attacks. Attempt to use IP fragmentation option to create extremely small fragments and force TCP header information into a separate packet fragment. IT attempts to circumvent the filtering rules that depend on TCP header information.

Countermeasure: enforcing the rule that the first fragment of a packet must contain a predefined minimum amount of the transport header. If the first fragment is rejected, then the filter can remember the packet and discard all subsequent fragments.

Stateful Inspection Firewalls

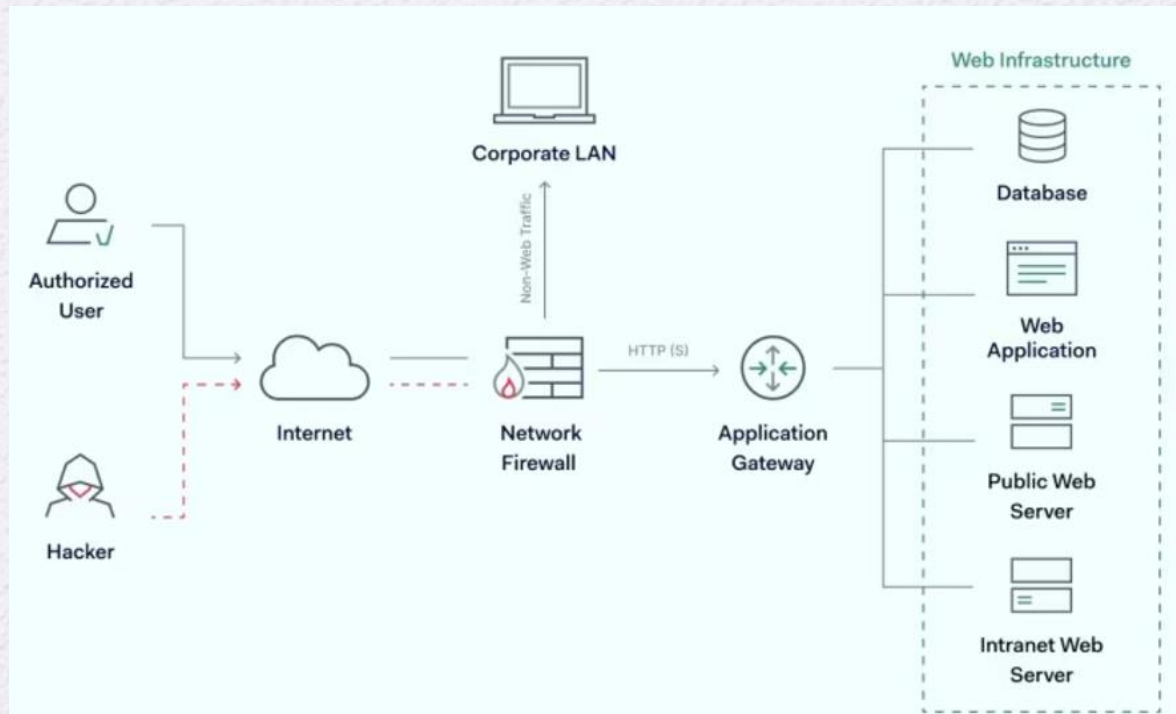
- A stateful packet inspection firewall reviews the same packet information as a **packet filtering firewall**, but **also records information about TCP connections**.
- The firewall compares incoming packets with a state table that contains information about current TCP connections.

Table 21.1 Example Stateful Firewall Connection State Table

Source Address	Source Port	Destination Address	Destination Port	Connection State
192.168.1.100	1030	210.9.88.29	80	Established
192.168.1.102	1031	216.32.42.123	80	Established
192.168.1.101	1033	173.66.32.122	25	Established
192.168.1.106	1035	177.231.32.12	79	Established
223.43.21.231	1990	192.168.1.6	80	Established
219.22.123.32	2112	192.168.1.6	80	Established
210.99.212.18	3321	192.168.1.6	80	Established
24.102.32.23	1025	192.168.1.6	80	Established
223.21.22.12	1046	192.168.1.6	80	Established

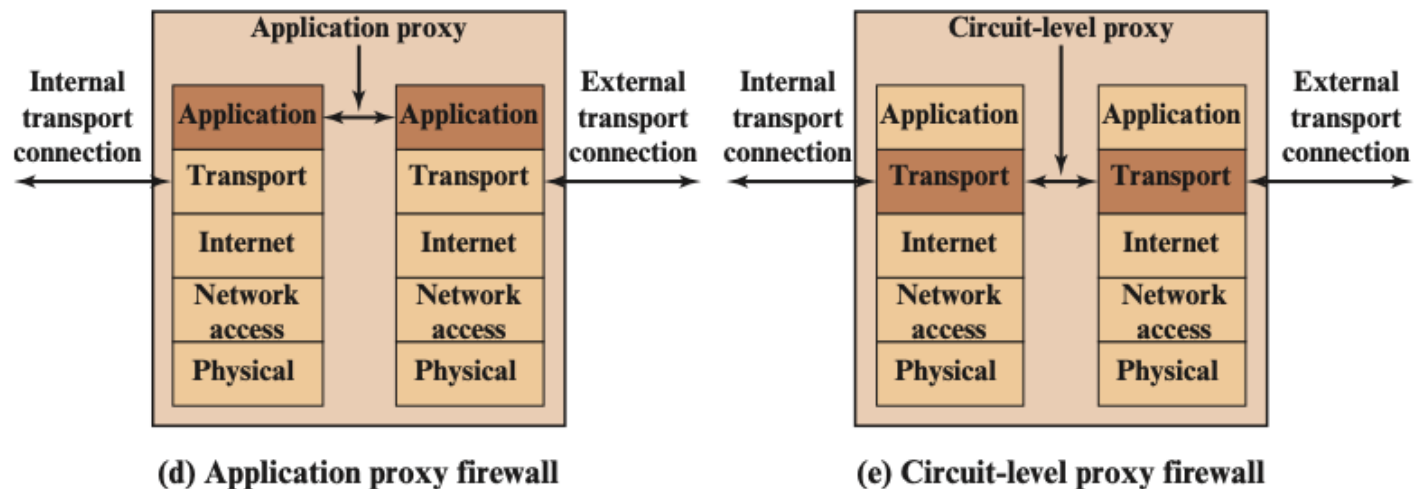
Application-Level Gateway (Proxy Firewall)

- An application-level gateway, **also known as an application proxy**, operates as a relay for application-level traffic, here how it works
 - **Client Interaction:** The **user initiates a connection** using a TCP/IP protocol using an application (like Telnet or FTP) and **contacts the firewall gateway**.
 - **Authentication:** Upon connection, the gateway asks the user for information such as the remote host's name, user ID, and **authentication credentials**.
 - **Relaying Traffic:** The gateway then contacts the application on the remote host, **forwarding the application data between the client and the server**.
- **Application-Specific Proxy:** The gateway **only relays traffic** for applications it has been **specifically configured to handle**.
- **Feature Control:** Administrators can configure the gateway to allow only **certain features of an application**.
- **Prime disadvantage:** the additional processing overhead on each connection gateway must examine all traffic in both directions.



Circuit-level gateway

- Circuit-level gateway **does not** permit direct TCP connections;
- The circuit-level gateway **establishes two separate TCP connections**: from client (external) to firewall, and from firewall to server (internal). Then it **acts as a relay**.
- It simply forwards TCP segments (i.e., packets) from one connection to the other without inspecting the contents of the traffic.
- A typical use case for a circuit-level gateway is when a network administrator **trusts the external users** but wants to **control the connections** that are established from the internal network to the external network (and vice versa).



Demilitarized Zone (DMZ)

- Positioned between external and internal firewalls.
- The **DMZ hosts publicly accessible** systems like web servers, email servers, and DNS servers. These systems require external connectivity (e.g., to the internet) but also need protection from the internet.
- **External Firewall:** Provides **basic** protection and access control for the DMZ and the enterprise network from the external network.
- **Internal Firewalls:** Offer **more detailed filtering** and protect both the core network and the DMZ, ensuring that internal systems are protected from external attacks
- **Common practice:** place the DMZ on a different network interface on the external firewall from that used to access the internal networks.
- **Benefits**
 - Layered defense
 - Controlled exposure
 - Internal Segmentation

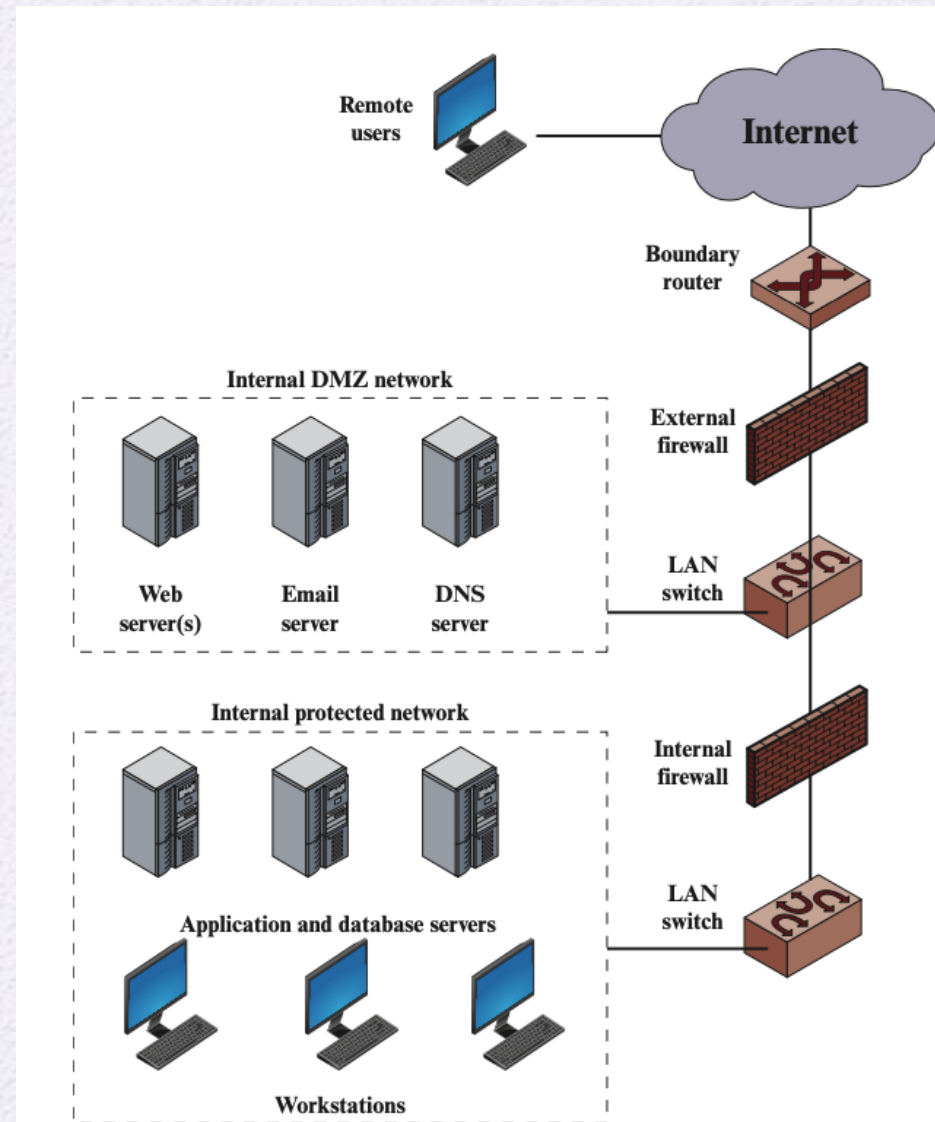


Figure 21.3 Example Firewall Configuration

Intrusion Detection Systems (IDS)

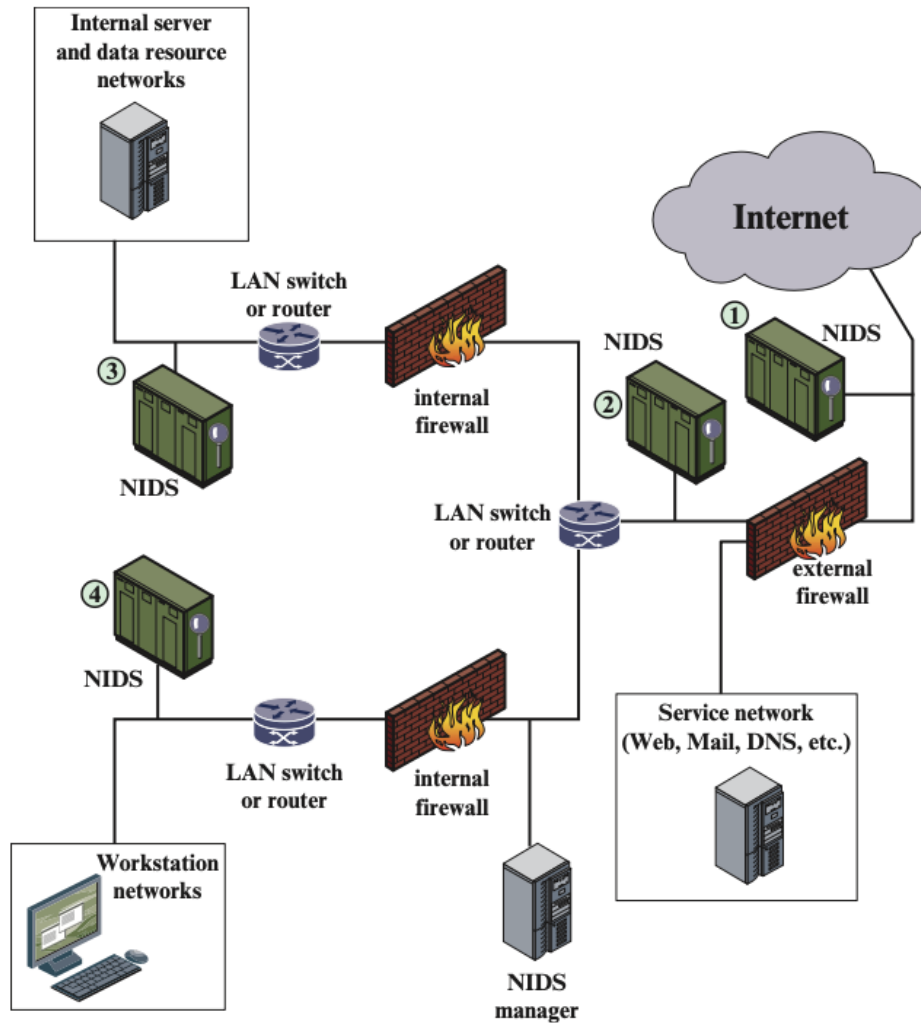


Figure 21.6 Example of NIDS Sensor Deployment

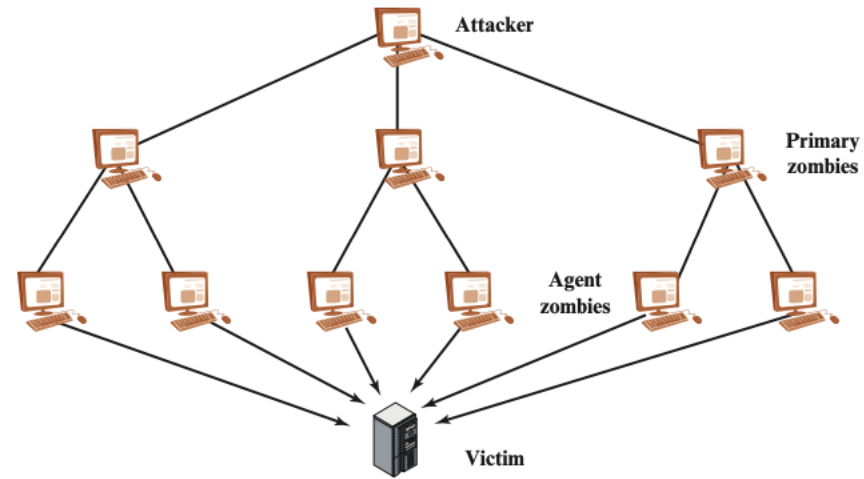
Host-Based Intrusion Detection Techniques Network-Based Intrusion Detection Systems



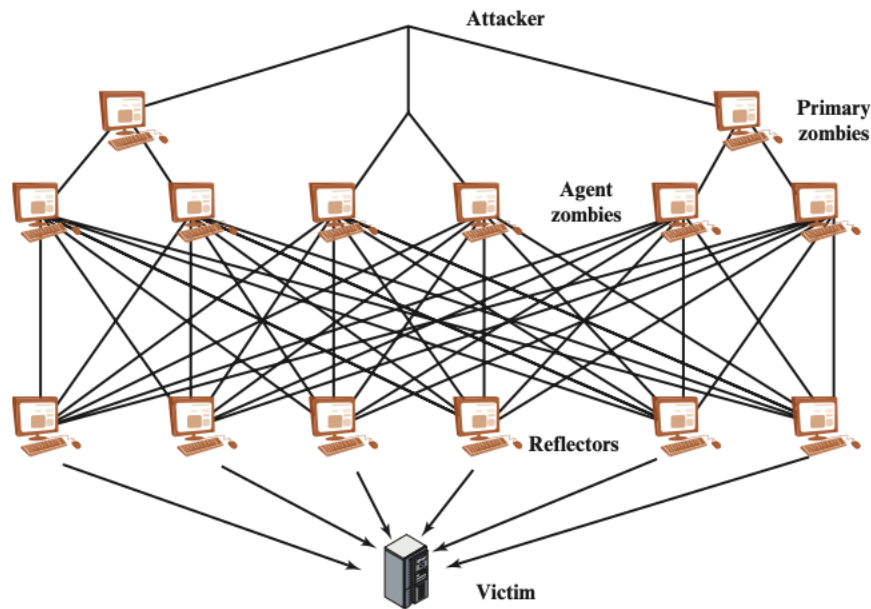
		Area of Vulnerability		
		Network	Payload	Endpoint
Time Scale	Real-Time/ Near-Real-Time	Network Traffic Analysis	Payload Analysis	Endpoint Behavior Analysis
	Post-compromise (days/weeks)	Incident Management and Forensics		

Figure 21.7 Five Elements of Malware Defense

Denial of Service Attacks (DDoS)



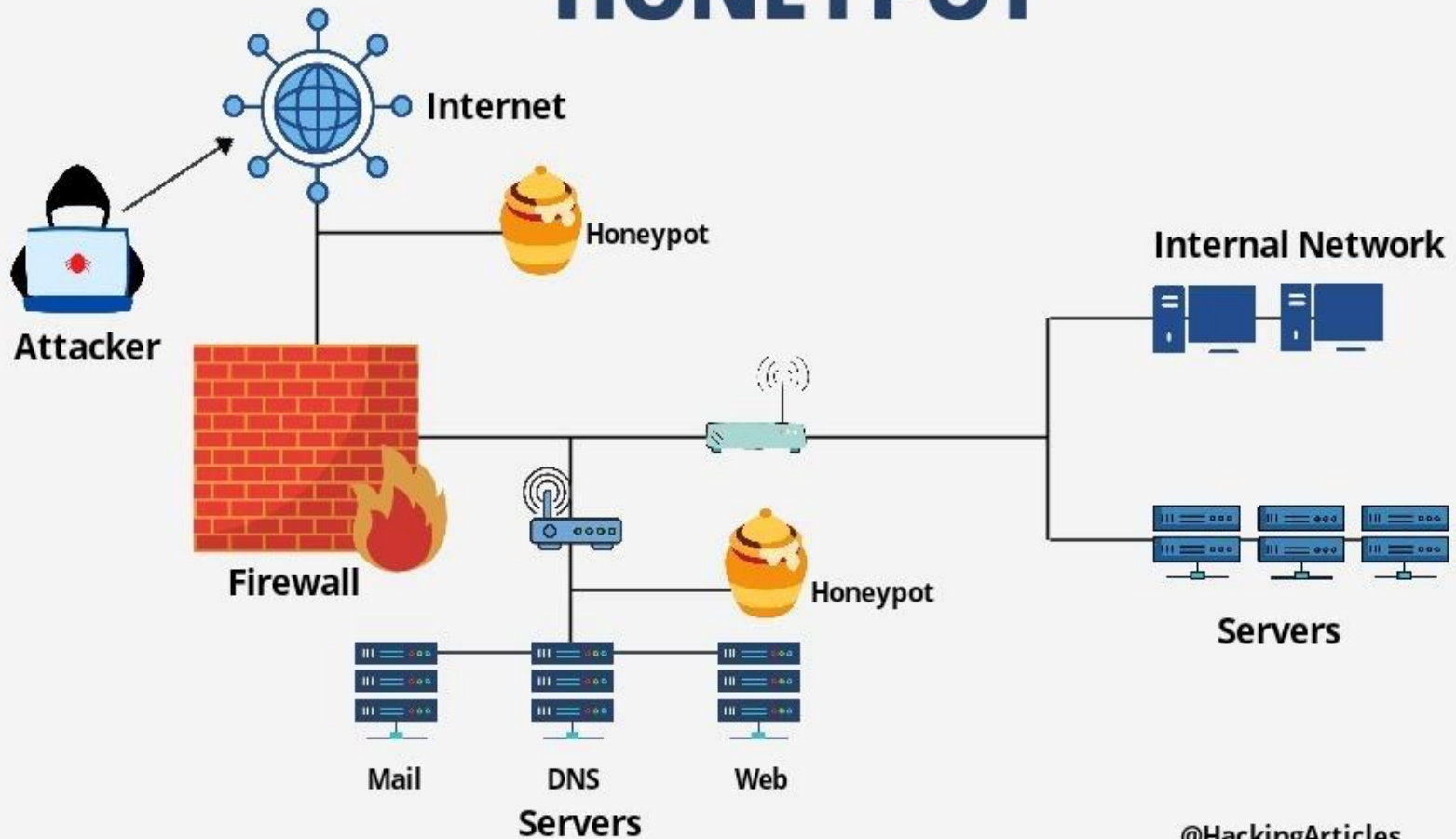
(a) Direct DDoS attack



(b) Reflector DDoS attack

Figure 21.9 Types of Flooding-Based DDoS Attacks

HONEYPOT



@HackingArticles

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